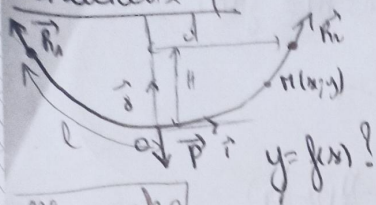
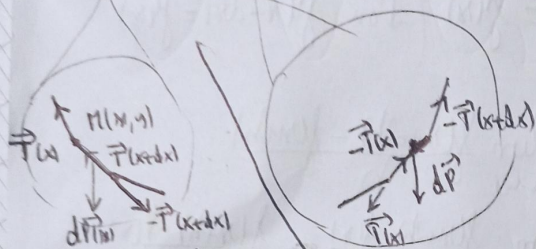
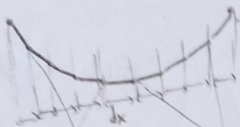


Chaîne de L



une approche



En appliquant le TCI sur le fragment du milieu, on a:

$$dT_x + T(x) - T(x+dx) = 0 \Rightarrow dT_x = -dT_x$$

$$\text{càd } \begin{cases} dT_x(x) = 0 \\ dT_x(x) = -dP(x) = -\mu dx g \end{cases}$$

$$\text{càd } \begin{cases} T_x(x) = T_x \\ (s) \frac{dT_x(x)}{dx} = -\mu g \sqrt{1+y'(x)^2} \end{cases}$$

$$\text{ou } T_{y(x)} = y'(x-dx)$$

$$\text{d'où } T_y(x) = T_x y'(x-dx)$$

$$\text{d'où } \frac{dT_y(x)}{dx} = \frac{T_x(y'(x) - y'(x-dx))}{dx}$$

$$= T_x \lim_{h \rightarrow 0} \frac{y'(x) - y'(x-h)}{h}$$

$$= T_x \lim_{h \rightarrow 0} \frac{y'(x+h) - y'(x)}{h}$$

$$(\text{ou } h = -h \rightarrow 0)$$

$$= T_x y''(x)$$

$$\text{d'où } (s) \Rightarrow \begin{cases} T_x(x) = T_x(h) \\ \frac{y''(x)}{\sqrt{1+y'(x)^2}} = -\frac{\mu g}{T_x} \end{cases} \quad (2)$$

$$\text{càd } (2) \Rightarrow \frac{y''(x)}{\sqrt{1+y'(x)^2}} = -\frac{\mu g}{T_x} \quad (2)$$

$$\text{càd } (2) \Rightarrow \frac{y''(x)}{\sqrt{1+y'(x)^2}} = -\frac{\mu g}{T_x} \quad (2)$$

$$(2) \Rightarrow y'(x) = \frac{\mu g}{T_x} \left(-\frac{1}{\sqrt{1+y'(x)^2}} + C_1 \right)$$

$$\text{ou } y'(0) = 0 \Rightarrow \frac{\mu g}{T_x} \left(-\frac{1}{\sqrt{1+y'(0)^2}} + C_1 \right) = 0 \Rightarrow C_1 = 0$$

$$(2) \Rightarrow y'(x) = \frac{\mu g}{T_x} \left(-\frac{1}{\sqrt{1+y'(x)^2}} + C_2 \right)$$

$$\text{ou } y'(0) = 0 \Rightarrow C_2 = -1$$

$$(2) \Rightarrow y(x) = \frac{\mu g}{T_x} \left(-\frac{1}{\sqrt{1+y'(x)^2}} - 1 \right)$$

$$\text{ou } y(d) = H \Rightarrow \frac{\mu g}{T_x} \left(-\frac{1}{\sqrt{1+y'(d)^2}} - 1 \right) = H$$

$$\Rightarrow -\frac{\mu g}{T_x} = \ln(H+1 + \sqrt{(H+1)^2 - 1})$$

$$\Rightarrow T_x = \frac{\mu g d}{\ln(H+1 + \sqrt{(H+1)^2 - 1})}$$

$$\text{Donc } y(x) = \frac{\mu g}{T_x} \left(\frac{\ln(H+1 + \sqrt{(H+1)^2 - 1})}{d} x \right) - 1$$

* Longueur L

$$L = \int_{-d/2}^{d/2} dx \sqrt{1+y'(x)^2}$$

$$= \int_{-d/2}^{d/2} \sqrt{1 + \left(\frac{\mu g}{T_x} \right)^2 x^2} dx$$

* Tension T

$$\text{on a eu } T_x = \frac{\mu g d}{\ln(H+1 + \sqrt{(H+1)^2 - 1})}$$

$$dT_y(x) = T_x y'(x) = \frac{\mu g d}{\ln(H+1 + \sqrt{(H+1)^2 - 1})} \times \frac{\ln(H+1 + \sqrt{(H+1)^2 - 1})}{d} \ln(x)$$

$$\text{Donc } T = \frac{\mu g d}{\ln(H+1 + \sqrt{(H+1)^2 - 1})} T +$$

$$(2) \Rightarrow y(x) = \frac{1}{d} \ln(x) + C_2 \text{ ou } C_2 \text{ est tel que } -\frac{\mu g}{T_x}$$

$$y(0) = 0 \Rightarrow C_2 = -\frac{1}{d}$$

$$y(d) = H \Rightarrow \frac{1}{d} \ln(d) - \frac{1}{d} = H$$

c'est la solution de celle eqn. L'en m → 1/(ch(md)-1) la jecture de K et de B.

Donc $y(x) = \frac{1}{a} (\text{ch}(ax) - 1)$

où $a = -\frac{Mg}{T_x} \text{ et } T_x = T_0 \cdot i$

* longueur L

$$L = \int_{-l}^l dl = \int_{-l}^l \sqrt{1 + y'^2} dx$$

$$= \int_{-l}^l \sqrt{1 + \sinh^2(ax)} dx$$

$$= \int_{-l}^l \cosh(ax) dx$$

$$= \frac{1}{a} [\sinh(ax)]_{-l}^l$$

$$L = \frac{2}{a} (\sinh(al))$$

En connaissant L on a:

$$\sinh(al) = \frac{aL}{2}$$

$$\cosh(al) = \sqrt{1 + \left(\frac{aL}{2}\right)^2}$$

on $\cosh^2(al) - \sinh^2(al) = 1$ car

$$\left(\sqrt{1 + \left(\frac{aL}{2}\right)^2}\right)^2 - \left(\frac{aL}{2}\right)^2 = 1$$

~~on a~~

où $a = \frac{2H}{L^2 - H^2}$ ou $L = \frac{L}{2}$

$$\text{car } y(x) = \frac{L^2 - H^2}{2H} \left(\cosh\left(\frac{2H}{L^2 - H^2} x\right) - 1 \right)$$

* Tension

car $a = -\frac{Mg}{T_x}$ on déduit que

$$T_x = -\frac{Mg(L^2 - H^2)}{2H} \quad \text{et aussi que}$$

$$\vec{T} = -\frac{Mg(L^2 - H^2)}{2H} \left(\vec{i} + \sinh\left(\frac{2H}{L^2 - H^2} x\right) \vec{j} \right)$$

Note: $f'(x \pm dx) = \lim_{h \rightarrow 0} f'(x \pm h) = f'(x)$

$$\frac{df(x \pm dx)}{dx} = \frac{f(x \pm h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(x) - f(x-h)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

où $h = -h$

$$\frac{df(x-dx)}{dx} = f'(x) \quad \text{donc } f'(x-dx) = f'(x)$$

$$\frac{df(x+dx)}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

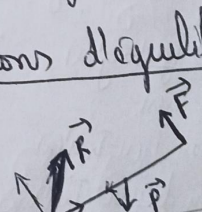
$$= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} + \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2(f(x+h) - f(x))}{h} = 2f'(x)$$

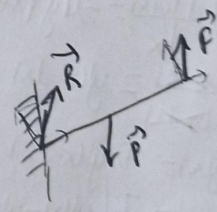
$$= 2f'(x)$$

* Notions d'équilibre

cas 1:



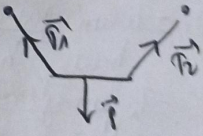
cas 2:



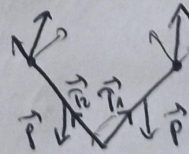
cas 3:



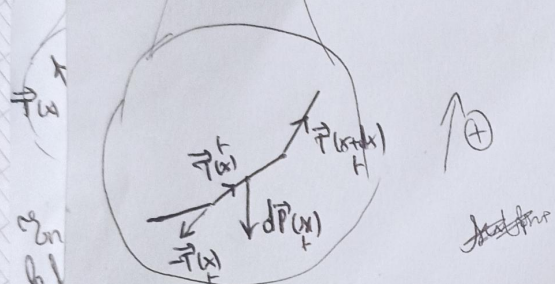
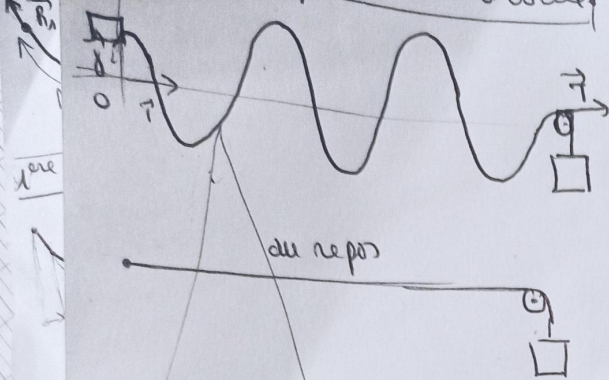
cas 4:



cas 5:



ondes mécaniques : cas de la corde



En appliquant le P.T.I sur le fragment du milieu,

$$\vec{dP}(x) + \vec{T}(x+dx) - \vec{T}(x) = dm \vec{a}(x)$$

car $\vec{dP}(x) + \vec{T}(x) = dm \vec{a}(x)$ on le met sous

$$\left. \begin{aligned} dT_x(x) &= 0 \\ dP(x,t) + dT_y(x,t) &= dm \ddot{y}(x,t) \end{aligned} \right\}$$

$$\left. \begin{aligned} T_x(x,t) &= T_x(x) \quad (\text{on néglige } dP \text{ ou du} \\ dT_y(x,t) &= \mu dx \ddot{y}(x,t) \quad (\text{moins on va en} \\ &\quad \text{apertubeur}) \end{aligned} \right\}$$

$dm = \mu dx$ et non μdl car dl est la longueur du fil étiré et non celle au repos qui est dx

$$\text{on } \frac{T_y(x,t)}{T_x(x,t)} = \ddot{y}(x,t) \text{ d'où } \frac{\partial T_y(x,t)}{\partial x} = T_x \ddot{y}(x,t)$$

$$(2) \Rightarrow \ddot{y}(x,t) = \frac{\mu}{T_x} \ddot{y}(x,t)$$

$$\Rightarrow \left[\frac{\partial^2 y(x,t)}{\partial x^2} - \frac{\mu}{T_x} \frac{\partial^2 y(x,t)}{\partial t^2} \right] = 0 \quad (1)$$

Quand le système est immobile $T_x(x,t) = T$ d'où $T_x = T$

$$(1) \Rightarrow y(x) = A \sin \left(\frac{2\pi}{\lambda} x + \phi \right)$$

$$(1) \Rightarrow \left(\frac{\partial}{\partial x} - \frac{\partial}{\partial t} \right) \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial t} \right) y = 0 \text{ soit } \frac{\partial}{\partial t} \frac{\partial y}{\partial x} = 0$$

$$\text{on a en fait } \frac{\partial}{\partial x} \times \frac{\partial}{\partial t} y = 0 \quad (2)$$

$$\text{d'où } \frac{\partial}{\partial x} = \frac{\partial}{\partial x} - a \frac{\partial}{\partial t} \text{ et } \frac{\partial}{\partial t} = \frac{\partial}{\partial t} + a \frac{\partial}{\partial x}$$

il suffit d'avoir $p = x - at$ et $q = x + at$

$$\text{avec } (1) \Rightarrow \frac{\partial}{\partial x} \left(\frac{\partial}{\partial t} y \right) = 0$$

$$\Rightarrow \frac{\partial}{\partial x} y(p,q) = C_1(q)$$

$$\Rightarrow y(p,q) = \int C_1(q) dq + g(p)$$

$$\text{Donc } y(x,t) = f(x - \sqrt{\frac{\mu}{T}} t) + g(x + \sqrt{\frac{\mu}{T}} t)$$

$$\text{Pour une vibration sinusoidale, } y(x,t) = y_m \cos \left(b \left(x - \sqrt{\frac{\mu}{T}} t \right) \right) \text{ où } [b] [x] = 1$$

$$= y_m \cos \left(b \sqrt{\frac{\mu}{T}} t - b x \right)$$

on a une période temporelle $\frac{2\pi}{b \sqrt{\frac{\mu}{T}}} = \frac{2\pi}{b \sqrt{\frac{\mu}{T}}}$ et spatiale $\frac{2\pi}{b}$ car

$$y(x+\lambda, t) = y(x, t) \text{ et } y(x, t+T) = y(x, t)$$

on a donc à faire à une onde qui se déplace à la vitesse $v = \frac{\lambda}{T} = \sqrt{\frac{T}{\mu}}$ et

on peut alors écrire

$$y(x,t) = y_m \cos \left(\frac{2\pi}{\lambda} x - \frac{2\pi}{T} t + \phi \right)$$

$$\lambda = vT$$